BUSINESS CASE

Customer Churn — Retail Banking Portfolio

Source: Customer_Churn_Dataset | Prepared as a Business Analyst

1. Executive Summary

Overall churn stands at 20.4% (204 of 1,000 customers). Demographic factors (age, gender) show almost no difference in churn risk, which rules them out as levers. The one segment with a materially higher churn rate is customers classified as "LOW" on login-frequency risk 35.1% churn versus 17.7–19.1% for the other two risk bands. Married customers (23.0%) and low-income customers (22.2%) also churn somewhat above average. Spend levels and service-resolution outcomes are not meaningfully different between churned and retained customers, so this is not primarily a service complaint or a big-spender problem it looks like a disengagement problem.

Recommendation: launch a re-engagement program targeted at the 10% of customers falling into the low-frequency-risk band, rather than a broad, unfocused retention campaign.

2. Business Problem

The bank wants to reduce customer attrition but currently applies retention efforts broadly, without knowing which customer segments actually drive churn. Undirected retention spend is inefficient and dilutes the offers that would matter most to at-risk customers.

3. Data & Method

Four linked tables from the dataset were used:

- Customer_Demographics (1,000 customers): age, gender, marital status, income, churn status
- Customer_Service (1,002 interactions): interaction type and resolution status
- Online_Activity (1,000 records): login frequency and a derived frequency-risk band
- Transaction_History (5,054 transactions): amount spent and product category
- Each table was joined to Customer_Demographics on CustomerID, and churn rate was compared across segments within each table using cross-tabulation (the analysis equivalent of Excel COUNTIFS / pivot-table % of row).

4. Key Findings

4.1 Demographic drivers: weak signal

Segment	Churn Rate	Note
Married customers	23.0%	Highest of all marital-status groups
Divorced customers	18.5%	Lowest of all marital-status groups
Low income	22.2%	vs 19.2% for High income
Male	21.1%	vs 19.7% for Female — minor gap
Age group (Teen–Senior)	19.5% – 20.7%	Essentially flat — age is not a driver

4.2 Engagement: the strongest signal found

Login-Frequency Risk Band	Customers (n)	Churn Rate
LOW	97	35.1%
HIGH	717	19.1%
MEDIUM	186	17.7%

Customers in the "LOW" frequency-risk band churn at nearly double the rate of every other group, and this is not a small sample fluke 97 customers, 34 of whom churned. Average login frequency is also lower for churned customers overall (23.6 vs 26.5 logins), consistent with disengagement preceding churn.

4.3 Ruled out as primary drivers

- Service resolution: 20.5% churn after a resolved case vs 20.7% after an unresolved one no meaningful gap. Support ticket outcomes are not the main churn driver here.
- Spend: churned customers spent slightly more on average (₦253 vs ₦250 per transaction) this is not a low-value customer problem.
- Product category: Electronics buyers churn somewhat more (22.5%) than Books/Clothing buyers (18.7–18.9%), a secondary, smaller effect.

5. Root Cause

The data points to disengagement, not dissatisfaction, as the leading indicator of churn: customers who log in less often are the ones most likely to leave, while support-ticket outcomes and spend levels barely move the needle. Marital status and income show a mild compounding effect on top of this, suggesting the low-engagement segment may also skew toward married, lower-income customers who could be under-served by current digital engagement touchpoints.

6. Recommendation

1. Build a low-frequency-risk watchlist flag the 10% of customers in the LOW login-frequency band for proactive outreach before they churn, rather than waiting for a complaint or service ticket.
2. Test a re-engagement nudge (app notification, personalized email, or a relationship-manager call) targeted specifically at this segment, and track whether login frequency and churn improve over the following quarter.
3. De-prioritize service-resolution-based retention triggers as a primary lever the data does not support them as the main driver here and redirect that budget toward the engagement program above.
4. Layer in marital status and income only as secondary targeting criteria once the core engagement program is running, since their effect is real but smaller.

7. How to Validate This Before Acting

- A/B test the re-engagement nudge on a sample of the LOW-risk segment against a holdout group before a full rollout.
- Re-check this analysis next quarter to confirm the login-frequency effect holds as new data comes in, not just as a onetime pattern.
- Confirm with the online/digital team how "Frequency Risk" bands are actually defined, to make sure the recommendation targets the right underlying behaviour.